CENG424 Embedded Computer Systems
Spring 2026
HOMEWORK 4 - Servo Motor Position Control

Objective:

The goal of this assignment is to design a PID controller for controlling the position of a servo motor. You will simulate the system in Simulink, tune the PID controller, and evaluate its performance in terms of position accuracy and response time.

System Description:

In this system, we have a servo motor that controls the position of a rotating arm. The motor is driven by an input voltage $\mathrm{u(t)}$, which affects the position $\theta(\mathrm{t})$ of the arm. The dynamics of the servo motor can be described by the following second-order differential equation:

$$\mathrm{J}\frac{\mathrm{d}^2\theta(\mathrm{t})}{dt^2} + \mathrm{B}\frac{\mathrm{d}\theta(\mathrm{t})}{dt} = \mathrm{K_t}\,.\,\mathrm{u(t)}$$

Where:

- $\theta(\mathrm{t})$: Position of the motor (radians)
- $\mathrm{u(t)}$: Input control signal (voltage) to the motor
- J: Moment of inertia of the motor (kg·m²)
- B: Damping coefficient (N·m·s/rad)
- $\mathrm{K_t}$: Torque constant (N·m/A)

System Goal:

The goal is to control the position of the arm by adjusting the input voltage $\mathrm{u(t)}$ using a PID controller. You will aim to reach and maintain a desired position $\theta_{\mathrm{desired}}$ of the motor.

Steps:

1. Model the Servo Motor System in Simulink:

- Begin by representing the second-order differential equation in Simulink using blocks like Integrator, Gain, and Sum.
- You need to implement the equation
- J Represent $\theta(\mathrm{t})$ as the output and $\mathrm{u(t)}$ as the input control signal.

2. Design the PID Controller:

- Implement a PID controller that takes the position error $\mathrm{e(t)} = \theta_{\mathrm{desired}} - \theta(\mathrm{t})$ as input and generates the control signal $\mathrm{u(t)}$ to adjust the position.
- You can use the PID Controller block in Simulink for this purpose. Also, you can adjust PID parameters using the tune button.

3. Simulate the System:

To simulate, use the following parameter values for the servo motor system:

- Moment of inertia (J): 0.005 kg·m²
- Damping coefficient (B): 0.6 N·m·s/rad
- Torque constant (K_t): 0.08 N·m/A
- Desired position ($\theta_{desired}$): 180° or π radians
- Initial motor position $\theta(0)$: 0 radians

Run the simulation and observe the motor's position response to the PID control. Simulation time should be sufficient to observe the motor's behavior.

4. Report the Results

- Your submission should include a script for the parameters, a Simulink model, and a report.
- Your report must include the system's Simulink model. Additionally, include the plots from the Scope blocks and evaluate the system's performance in terms of position accuracy and response time. **Please justify your answers in a few sentences.**